Year 8 Mathematics

Unit 1: Number

1.1 Calculating with Negative Integers

Calculating with Negative Integers

STARTER · 5 MIN

Date: _____ **Title: Calculating with Negative Integers**

LEARNING OBJECTIVES

- To be able to add and subtract positive and negative integers
- To be able to multiply and divide positive and negative integers

STARTER *Answer in your book. No calculators.*

1. Which is bigger: −12 or 9?
2. Which is smaller: −2 or −7?
3. Work out −4 + 8
4. Work out −2 − 4

Starter — Answers

STARTER

Tick or correct in green pen

1. **9**

2. **−7**

3. **4**

4. **−6**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can add or subtract when the signs are different	☐
SILVER	I can multiply or divide when the signs are different	☐
GOLD	I can apply integer arithmetic to solve real-world problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Calculating with Negative Integers

INTRODUCTION · 3 MIN

An integer is a whole number: positive, negative or zero.

Adding a negative

Two different signs together: −

$3 + -10 = 3 - 10 = -7$

Subtracting a negative

Two signs the same together: +

$-13 - -9 = -13 + 9 = -4$

× and ÷: same signs

The answer is positive.

$-1 \times -9 = 9$ $\quad 15 \div 3 = 5$

× and ÷: different signs

The answer is negative.

$3 \times -10 = -30$ $\quad -4 \div 2 = -2$

Examples 1 and 2 — Adding and Subtracting on a Number Line

EXAMPLES · 5 MIN

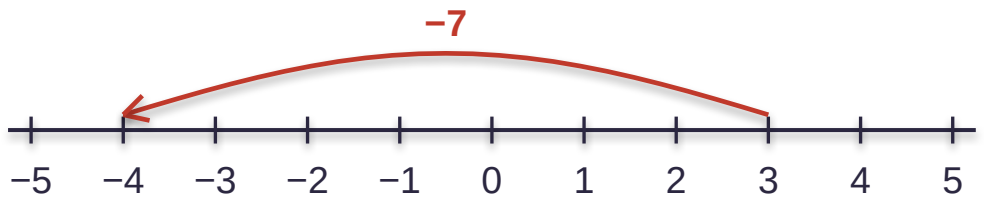
BRONZE

Watch and copy into your book

Hint + moves right − moves left

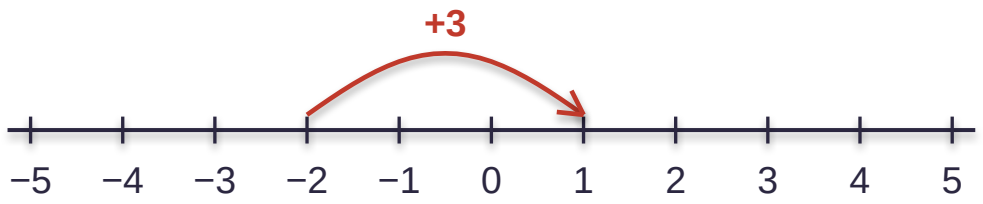
Example 1

Work out 3 − 7



Example 2

Work out −2 + 3



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Start at 3, move 7 to the left

$$3 - 7 = -4$$

Tip: subtracting moves left.

Example 2

Start at −2, move 3 to the right

$$-2 + 3 = 1$$

Tip: adding moves right.

Task 1

PRACTICE · 9 MIN

BRONZE *Work out*

Hint + moves right − moves left

1. $2 - 6$
2. $-1 - 5$
3. $-9 + 3$
4. $-7 + 22$
5. $-5 + 5$

★ $4 - 11$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 **−4**

2 **−6**

3 **−6**

4 **15**

5 **0**

★ **−7**

Examples 3 and 4 — Two Signs Together and × ÷

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint Same signs: + Different signs: −

Example 3

Work out

(a) $3 + -10$

(b) $-13 - -9$

Example 4

Work out

(a) 3×-10

(b) -1×-9

(c) $-4 \div 2$

(d) $-24 \div -6$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Different signs: − **Same signs: +**

(a) $3 + -10$ $= 3 - 10$

$= -7$

(b) $-13 - -9$ $= -13 + 9$

$= -4$

Tip: replace the two signs with one.

Example 4

Same signs: + **Different signs: −**

(a) 3×-10 $= -30$

(b) -1×-9 $= 9$

(c) $-4 \div 2$ $= -2$

(d) $-24 \div -6$ $= 4$

Tip: work out the number, then the sign.

Task 2

PRACTICE · 9 MIN

SILVER *Work out*

Hint Same signs: + Different signs: −

1. $-9 + -5$
2. $15 - -2$
3. $10 + -12$
4. 2×-1
5. -8×2

★ $-45 \div -9$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **−14**
2. **17**
3. **−2**
4. **−2**
5. **−16**
★ **5**

Examples 5 and 6 — Several Steps and Real Life

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint × ÷ before + − Rose: + Fell: −

Example 5

(2 marks)

Work out

$-6 + 12 \times -2$

Example 6

(1 mark)

At 6 am it was −4 °C. The temperature dropped 7 degrees.

What is the new temperature?

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Multiply first, then add

$12 \times -2 = -24$

$-6 + -24 = -6 - 24$

$= \mathbf{-30}$

Tip: × and ÷ before + and −.

Example 6

Dropped means subtract

$\mathbf{-4 - 7 = -11\ °C}$

Tip: start at −4, move 7 down.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show your working

Hint × ÷ before + − Rose: + Fell: −

1 −5 + 8 − 3

2 12 − −6 − 15

3 40 ÷ −5 + 18

4 −100 ÷ −20 + 15

5 A lift is on floor −3. It goes up 8 floors. Which floor now?

★ Can a negative number squared be negative? Explain.

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. **0**

2. **3**

3. **10**

4. **20**

5. **Floor 5**

★ **No: $(-3)^2 = -3 \times -3 = 9$**
same signs give +

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Work out 9 − 12 and −3 − −9

SILVER Work out 9 × −7 and −10 × −5

GOLD It is −4 °C and it drops 7 degrees. New temperature?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **−3 and 6**

SILVER **−63 and 50**

GOLD **−4 − 7 = −11 °C**

Exam Practice — Adding and Subtracting

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint + moves right − moves left

Question 1

(2 marks)

Work out

(a) −6 + 10 (1)

.....

(b) −3 − 5 (1)

.....

Question 2

(2 marks)

Write these numbers in order of size.
Start with the smallest.

4 −7 0 −2 9 −10

.....

Exam Practice — Two Signs Together

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint Same signs: + Different signs: −

Question 3

(2 marks)

Work out

(a) 8 − −5 (1)

.....

(b) −7 + −6 (1)

.....

Question 4

(3 marks)

Work out

(a) −6 × 4 (1)

.....

(b) −5 × −8 (1)

.....

(c) −36 ÷ 9 (1)

.....

Exam Practice — Problems with Integers

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint × ÷ before + − Rose: + Fell: −

Question 5

(2 marks)

Work out $25 - 18 - -7$

.....

Question 6

(3 marks)

At midnight the temperature in Moscow was −8 °C.

By noon it had risen by 11 degrees.
In the evening it fell by 6 degrees.

(a) The temperature at noon? (1)

..... °C

(b) The evening temperature? (2)

..... °C

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 4 B1

(b) −8 B1

Tip: use a number line.

Question 2

−10, −7, −2, 0, 4, 9 B2

B1 for one error

Tip: −10 is further below 0 than −7.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $8 + 5 = 13$ — B1

(b) $-7 - 6 = -13$ — B1

Tip: same signs +, different −.

Question 4

(a) -24 — B1

(b) 40 — B1

(c) -4 — B1

Tip: two negatives multiply to a positive.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

25 − 18 = 7 — M1

7 + 7 = 14 — A1

Tip: − − 7 means + 7.

Question 6

(a) −8 + 11 = 3 °C — B1

(b) 3 − 6 = −3 °C — M1 A1

Tip: rose means add, fell means subtract.